

AC-02

Exact certificates and a component obstruction

Complex bilinear rank of the 3×3 matrix product

Research report and reproducible verification package

14 September 2026

Status of the original problem: not resolved by this report.

The exact minimum is not established. The verified general interval remains $19 \leq R_{\mathbb{C}}(M_3) \leq 23$. No rank-22 identity and no global lower bound of 23 are supplied. The results below concern precisely specified families and necessary conditions for arbitrary decompositions.

Main computer-assisted result. The complete coefficient variety with the displayed Laderman zero pattern is an algebraic torus of dimension 58. Its closure under the standard three matrix changes of basis is a 76-dimensional irreducible component of the 23-slot Brent variety. Every slot has a fixed nonzero polynomial signature on this component, so no zero-slot point lies on it.

Additional results. Exact support obstructions are verified for Laderman’s and Sun’s 23-product schemes. A reshuffling argument gives general matrix-rank-sum constraints and a sharper classification of the output-rank profile in a tight hypothetical 22-product decomposition.

The archive contains the proofs, exact integer and Laurent certificates, a replay checker, attributed algorithm data, rational family instances, and numerical exploration logs. The checking is not a Lean formalization and has not undergone an independent external mathematical audit. No claim of priority for the restricted results is made.

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1 The target and the result actually obtained

AC-02 asks for the least number of separated bilinear products needed to multiply arbitrary complex 3×3 matrices. Each product must multiply one linear form in the first input by one linear form in the second input; scalar linear combinations are free. This is the exact tensor-rank model, not an addition-count, approximate, border-rank, or mixed-input model [1].

The established bounds used here are

$$19 \leq R_{\mathbb{C}}(M_3) \leq 23. \quad (1)$$

Bläser’s survey records these bounds in its introduction and Example 5.10 [2]. The lower bound is imported from the literature, not reproved by this report. The upper bound is verified here from an explicit Laderman identity, and separately from Sun’s displayed identity [3, 4]. The source checks did not locate a qualifying improvement for the exact complex model; the repository’s current statement and Alman–Li also identify the exact-rank question as unresolved [1, 6].

Wang’s finite-field lower-bound work has a different field of coefficients [5]. In particular, a nonexistence result for coefficients in $\mathbb{F}_2$ does not establish nonexistence over $\mathbb{C}$. No finite-field search result is used here to infer a complex tensor-rank lower bound.

1.1 An exact separation of conclusions

The principal conclusions proved with the accompanying finite certificates are summarized below. A “slot” means one ordered rank-one tensor summand, including its three coefficient vectors.

Conclusion	Scope
All 729 identities hold for two 23-slot decompositions	Existing upper bounds
No factor vector can vanish with the prescribed supports	Laderman and Sun supports
All three pairwise Khatri–Rao matrices have column rank 23	The same support varieties
The complete support variety is $(\mathbb{C}^*)^{58}$	Laderman support
A 76-dimensional component contains no zero-slot point	The component through Laderman
$\sum_t \text{rank } W_t \geq 27$, and analogues for U, V	Every exact decomposition
A tight 22-slot output profile is $2^5 1^{17}$	Conditional on $\sum_t \text{rank } W_t = 27$

These conclusions neither exhibit a 22-product algorithm nor exclude all such algorithms. In particular, the component theorem does not assert that every decomposition is equivalent to Laderman’s. The failure of that global coverage statement is the central gap, not a missing computational check within the displayed component.

2 Coordinates, identities, and the coefficient variety

Throughout, matrix indices i, j, k range over $\{0, 1, 2\}$. A matrix entry (i, j) has vector coordinate $3i + j$. Product slots are indexed $t = 1, \dots, r$ in the text and $t = 0, \dots, r - 1$ in JSON and Python unless explicitly labeled otherwise.

Write U_t, V_t, W_t for 3×3 coefficient matrices and define

$$AB = \sum_{t=1}^r \left(\sum_{i,j} U_{t,ij} A_{ij} \right) \left(\sum_{j,k} V_{t,jk} B_{jk} \right) W_t. \quad (2)$$

There is no complex conjugation in this formula or in any algebraic signature below. Equality for every input pair is equivalent to the 729 Brent equations

$$\sum_{t=1}^r U_{t,ij} V_{t,j'k} W_{t,i'k'} = \delta_{ii'} \delta_{jj'} \delta_{kk'}. \quad (3)$$

Let $T \in \mathbb{Z}^{9 \times 9 \times 9}$ have entry 1 at $(3i + j, 3j + k, 3i + k)$ and 0 elsewhere. It has 27 entries equal to 1 and 702 equal to 0. Equation (3) is $T = \sum_t u_t \otimes v_t \otimes w_t$, with row-major vectorized factors.

Definition 2.1. X_{23} is the affine variety in $\mathbb{C}^{621}$ cut out by (3) with $r = 23$. All slots and all factors are ordered. Zero factor vectors are allowed. For a fixed baseline $F^{(0)} = (U, V, W)$, its support variety consists of the points in X_{23} at which every coordinate zero in $F^{(0)}$ remains zero. Permitted coordinates are not assumed nonzero in this definition.

A point has a zero slot exactly when at least one of its three vectors in that slot is zero: a tensor product of three nonzero vectors over a field is nonzero. Hence

$$R_{\mathbb{C}}(M_3) \leq 22 \iff X_{23} \text{ contains a zero-slot point.} \quad (4)$$

The forward implication pads an algorithm with zeros; the reverse implication deletes the zero slot. An algorithm of fewer than 22 products can first be padded to 22 products as well.

2.1 Data and baseline verification

The files `data/laderman23.json` and `data/sun23.json` contain three 23×9 signed-integer arrays. They are reconstructed from explicit scalar formulas by `code/build_data.py`. Laderman's formulas are also printed in Appendix A. Sun's data are transcribed from Section 5 of his paper [4]; the data are not an algorithm discovered in this work.

The checker evaluates every left side of (3) as an integer and compares it to the corresponding Kronecker delta. All 729 checks pass for each baseline. This proves exact identities, not merely agreement on random numerical inputs. Because the identities have integer coefficients, the displayed algorithms also remain identities under extension to $\mathbb{C}$. Their optimality does not follow from verification.

3 Nonvanishing on two prescribed support varieties

For each permitted coordinate, introduce a normalized variable $x_a = F_a / F_a^{(0)}$. Baseline coefficients are ± 1 , so the normalization is harmless even before nonvanishing has been established. Each restricted Brent equation is a sum of signed cubic monomials, possibly including the constant -1 .

Lemma 3.1 (Two-monomial propagation). *Suppose a polynomial equation over a field is $c_1 m_1 + c_2 m_2 = 0$, where c_1, c_2 are nonzero scalars and m_1, m_2 are monomials. If every variable in m_1 is nonzero at every solution of the system, then every variable in m_2 is also nonzero at every solution. The empty monomial 1 is nonzero without any assumptions.*

Proof. The first monomial is nonzero. The equation gives $m_2 = -c_1 m_1 / c_2 \neq 0$. A product of field elements is nonzero only if each factor is nonzero. Applying the argument successively preserves its universal quantifier over all solutions. $\square$

For example, the coefficient of $A_{00}B_{00}$ in output $(0,0)$, on Laderman’s support, gives

$$x_{U_{6,00}} x_{V_{6,00}} x_{W_{6,00}} = 1.$$

These three coordinates are therefore forced nonzero. Other equations propagate nonvanishing from already established coordinates.

Proposition 3.2 (Exact support obstruction). *On the Laderman support variety, every one of the 153 permitted coordinates is nonzero. On the Sun support variety, at least 162 of the 175 permitted coordinates are nonzero. In both varieties all 69 factor vectors are nonzero, and consequently every one of the 23 slots is nonzero.*

Proof. The finite traces in `proofs/laderman23_forcing.json` and `proofs/sun23_forcing.json` contain respectively 87 and 99 applications of Lemma 3.1. Each step specifies the Brent polynomial, the monomial already known to be nonzero, and the newly forced variables. The checker reconstructs the equations from the baseline data; it does not accept a polynomial supplied by the certificate. It verifies the hypotheses and conclusion of every step and then checks that each of the 69 factor vectors contains a forced coordinate. The resulting sets have sizes 153 and 162, respectively. $\square$

This does not assume that the original ± 1 entries stay ± 1 . The allowed entries may vary arbitrarily in $\mathbb{C}$, including attempted zeros. It is precisely the equations that prevent the stated coordinates from vanishing. For Sun, no claim is made that the remaining 13 permitted coordinates must be nonzero.

3.1 A stronger pairwise independence certificate

For two modes, define the 81×23 matrix

$$K_{UV}(:, t) = u_t \otimes v_t,$$

and define K_{VW} and K_{WU} similarly. The ordering of their 81 rows is lexicographic in the two vector coordinates.

Lemma 3.3 (Triangular peeling). *Suppose the columns of a matrix have a fixed permitted support. Assume there is an ordering of the columns and a row for each column such that, among the columns not previously removed, the selected row has only that column as a permitted nonzero, and its entry is known to be nonzero. Then the columns are linearly independent for every permitted assignment.*

Proof. In a linear dependence, the first selected row forces the coefficient of its unique surviving column to be zero. Remove that column from the relation and repeat. Every coefficient is zero. $\square$

Proposition 3.4. *For every solution in either support variety of Proposition 3.2, each of K_{UV}, K_{VW}, K_{WU} has column rank 23.*

Proof. The two `*_pairwise.json` certificates each give three lists of 23 row–column choices. The checker verifies the support uniqueness at each step and verifies that the two coordinates whose product is the selected entry belong to the forced-nonzero set. Lemma 3.3 applies to each list. $\square$

As one consequence, with the two input-factor arrays fixed at any such solution, the output-factor array is uniquely determined by the tensor identity: the matrix equation is $K_{UV}W = T_{UV|W}$ and K_{UV} is injective. Reoptimizing only the output factors therefore cannot remove a slot. This is an obstruction for the displayed pairs, not a lower bound against all alternative pairs of factors.

4 The complete Laderman support variety is a torus

Let $S \subseteq X_{23}$ denote the Laderman support variety. Proposition 3.2 places all its 153 allowed coordinates in $\mathbb{C}^*$. Thus its equations may be treated as Laurent polynomial equations in the normalized coordinates $x_1, \dots, x_{153}$.

4.1 Reduction to unit-monomial equations

Of the 729 restricted equations, 610 vanish identically. The other 119 have the following forms: 23 equations have one cubic and a nonzero constant; 92 have two cubics and no constant; and four have three cubics and a constant. The first 115 therefore become equations $x^a = 1$, with integer exponent vectors a .

Let $A_0 \in \mathbb{Z}^{115 \times 153}$ consist of these exponent rows. The certificate establishes $\text{rank } A_0 = 91$ and gives a 91×91 minor whose inverse is an integer matrix. It also gives $E_0 \in \mathbb{Z}^{153 \times 62}$ with an identity block on 62 free coordinates and $A_0 E_0 = 0$.

In each of the four remaining equations, two oppositely signed cubics have exponent difference in the integer row lattice generated by the selected rows of A_0 . Explicit integer combinations in the certificate prove these four lattice memberships. Therefore those two cubics agree on every solution of $x^{A_0} = 1$ and cancel. The survivor gives one further equation $x^a = 1$.

For clarity, the four affected coefficient coordinates are listed below. The entries in the first column are 0-based vector coordinates (a, b, c) ; slot numbers in the last column are 1-based. All factor entries in that column are normalized by their baseline signs.

(a, b, c)	Brent index $81a + 9b + c$	Surviving monomial
$(3, 1, 4)$	256	$x_{U_{5,10}} x_{V_{5,01}} x_{W_{5,11}}$
$(5, 8, 5)$	482	$x_{U_{18,12}} x_{V_{18,22}} x_{W_{18,12}}$
$(6, 2, 8)$	512	$x_{U_{9,20}} x_{V_{9,02}} x_{W_{9,22}}$
$(8, 7, 7)$	718	$x_{U_{15,22}} x_{V_{15,21}} x_{W_{15,21}}$

Let $A \in \mathbb{Z}^{119 \times 153}$ be A_0 with these four exponent rows appended. Its certified rank is 95, and a selected 95×95 minor has an integer inverse.

Lemma 4.1 (Unimodular monomial elimination). *Let $A \in \mathbb{Z}^{m \times n}$. Select r rows and r pivot columns so that the resulting square matrix B has an integer inverse. Let C be the selected rows restricted to the other $n - r$ columns. Define $E \in \mathbb{Z}^{n \times (n-r)}$ by*

$$E_{\text{free},:} = I_{n-r}, \quad E_{\text{pivot},:} = -B^{-1}C.$$

If $AE = 0$, then the solution set of $x^A = 1$ in $(\mathbb{C}^)^n$ is isomorphic to $(\mathbb{C}^*)^{n-r}$ by $x_a = \prod_j z_j^{E_{aj}}$.*

Proof. For every z , the equations hold because their exponent vectors become zero after multiplication by E . Conversely, let z be the free coordinates of a solution. The selected equations are $x_{\text{pivot}}^B z^C = 1$.

Taking the integer monomial combinations specified by B^{-1} expresses each pivot variable uniquely as the corresponding row of $-B^{-1}C$. No extraction of roots is involved. All other equations hold for the parametrization by $AE = 0$. Both the map and its inverse, coordinate projection to z , are regular maps of tori. $\square$

Theorem 4.2 (Complete fixed-support parametrization). *There is an explicit isomorphism*

$$S \simeq (\mathbb{C}^*)^{58}.$$

More precisely, let E be the 153×58 integer matrix in `proofs/laderman_torus.json`, under `final.E`. Then every point of S , and only such a point, is

$$F_a(z) = F_a^{(0)} \prod_{j=1}^{58} z_j^{E_{aj}}, \quad z \in (\mathbb{C}^*)^{58}, \quad (5)$$

with all other coordinates zero. Every entry of E has absolute value at most 2.

Proof. Nonvanishing follows from Proposition 3.2. The preceding integer row-lattice certificates reduce the restricted equations to $x^A = 1$. The certificate contains the selected rows and pivot columns, the integer inverse, the 58 free indices, and E . The replay checker verifies both inverse identities, the identity block on the free coordinates, the pivot formula in Lemma 4.1, and $AE = 0$. That lemma proves completeness and uniqueness.

As a redundant check of sufficiency, the checker substitutes (5) into each of the 729 original equations, groups terms by their exact 58-dimensional exponent vectors, and verifies that every resulting integer coefficient is zero. Thus no ignored residual equation or numerical approximation enters the parametrization. $\square$

The integer inverse is important. A rational rank computation alone would not exclude additional finite components, such as those arising from $x^2 = 1$. Here the selected minors are unimodular, so those extra root choices do not occur.

4.2 A six-parameter slice after elementary rescalings

There are 46 independent term-rescaling parameters:

$$U_t \mapsto \lambda_t U_t, \quad V_t \mapsto \mu_t V_t, \quad W_t \mapsto (\lambda_t \mu_t)^{-1} W_t.$$

Also allow diagonal basis changes $P = \text{diag}(p_1, p_2, 1)$, $Q = \text{diag}(q_1, q_2, 1)$, $R = \text{diag}(r_1, r_2, 1)$ in the action of Section 5. These give a 52-parameter torus of support-preserving transformations.

Proposition 4.3 (Explicit elementary-gauge slice). *Every point of S has a unique representative, under this 52-parameter rescaling group, for which 52 designated free coordinates in (5) equal 1. The other six free coordinates have 0-based indices*

$$21, 37, 45, 47, 50, 52.$$

The representative is obtained by retaining precisely these six columns of E and setting all other parameters to 1.

Proof. Let B be the 153×55 exponent matrix for all 46 term rescalings and all nine diagonal entries of P, Q, R . Let D be its restriction to the 58 free coordinate rows. Then $ED = B$; this identity is checked over $\mathbb{Z}$.

Select the 52 columns corresponding to all term rescalings and the first two diagonal entries of each matrix. The certificate gives 52 rows of D whose square submatrix has an integer inverse. These rows are exactly the complement of the six displayed indices. For any free-coordinate vector z , the monomial equations making these 52 coordinates equal to 1 have a unique solution for the 52 rescaling parameters, by the same integer-inverse argument as Lemma 4.1. The remaining three diagonal generators are integer monomial combinations of the selected generators on S , as is also verified in the certificate. $\square$

This is a quotient by the stated elementary rescalings, not a claim that the six coordinates classify algorithms under every possible equivalence. The file `data/laderman_six_parameter_exponents.csv` prints all 153 coefficient formulas as baseline signs and six exponent columns. The script `code/instantiate_family.py` creates exact rational instances. The sample with parameters $(2, 3, 5, 7, 11, 13)$ passes all 729 identities.

Parameterized matrix multiplication algorithms and heuristic ways of introducing parameters have substantial prior literature. In particular, Heule–Kauers–Seidl report that their particular iterative linear parameter-introduction procedure found no parameters for Laderman’s scheme [9, Section 7]. That is a statement about the output of a specified heuristic, not a theorem that every coupled nonlinear coefficient deformation is trivial. No priority claim is made here for the present torus calculation.

5 A 76-dimensional component without zero slots

The support theorem is not confined to algorithms that keep their zero patterns after a basis change. We now take an orbit closure inside the full coefficient variety X_{23} .

5.1 The action and a polynomial invariant

Let $G = \mathrm{GL}_3(\mathbb{C})^3$. Its action on coefficient matrices is

$$\begin{aligned} U'_t &= P^\top U_t Q^{-\top}, \\ V'_t &= Q^\top V_t R^{-\top}, \\ W'_t &= P^{-1} W_t R. \end{aligned} \tag{6}$$

It preserves (2): apply the original algorithm to $A' = PAQ^{-1}$ and $B' = QBR^{-1}$, and transform its output by $P^{-1}(\cdot)R$. In particular, it preserves X_{23} .

For a slot define the polynomial

$$\sigma_t(F) = \mathrm{tr}(U_t V_t W_t^\top) = \sum_{i,j,k} U_{t,ij} V_{t,jk} W_{t,ik}. \tag{7}$$

This is the familiar contraction signature in the theory of matrix multiplication decompositions [7]. Its invariance can be checked directly here:

$$U'_t V'_t (W'_t)^\top = P^\top U_t V_t W_t^\top P^{-\top}.$$

Taking trace proves $\sigma_t(F') = \sigma_t(F)$.

Proposition 5.1 (Constant signatures on the torus). *On S the signatures are*

$$\sigma_t = \begin{cases} 2, & t \in \{4, 7, 12, 16\}, \\ 1, & \text{all other slots.} \end{cases} \quad (8)$$

The slot numbering is that of the included Laderman data.

Proof. Substitute (5) in (7). The checker groups the resulting Laurent monomials by exponent vector. Every nonconstant coefficient is zero, and the constant coefficients are exactly (8). This holds symbolically for all 58 nonzero complex parameters, not just at the baseline point. $\square$

Define

$$Z = \overline{G \cdot S}^{\text{Zar}} \subseteq X_{23}, \quad (9)$$

where the closure is in affine coefficient space.

Theorem 5.2 (Orbit-closure obstruction). *Z is irreducible, and no point of Z has a zero slot.*

Proof. Both G and S are irreducible, so the closure of the image of the regular action map $G \times S \rightarrow X_{23}$ is irreducible. Every polynomial $\sigma_t - c_t$, with c_t from (8), vanishes on $G \cdot S$ and hence on its Zariski closure. At a zero slot, its cubic signature would be zero. This contradicts $c_t \in \{1, 2\}$. $\square$

This theorem also covers all finite coordinatewise limits of transformed family members. It does not cover limits in which the coefficient arrays diverge while only their summed tensor converges.

5.2 Exact local dimension and identification of the component

Let $L = F^{(0)}$ be the baseline Laderman point. Let J be the Jacobian of the 729 Brent polynomials at L , with columns ordered by mode, slot, then row-major entry. It is an integer 729×621 matrix. For example, the column for the variable $U_{t,a}$ is $e_a \otimes v_t \otimes w_t$, vectorized in lexicographic order.

Differentiate (5) at $z = (1, \dots, 1)$. Its 58 derivative columns have entry $F_a^{(0)} E_{aj}$ at each allowed coordinate and zero at the forbidden coordinates. Add 18 derivatives of (6), one for each off-diagonal elementary matrix D in each group factor:

$$\begin{array}{c|ccc} & \delta U_t & \delta V_t & \delta W_t \\ \hline P & D^\top U_t & 0 & -DW_t \\ Q & -U_t D^\top & D^\top V_t & 0 \\ R & 0 & -V_t D^\top & W_t D \end{array} \quad (10)$$

All derivatives are taken at the identity. Denote the resulting 621×76 matrix by H .

Lemma 5.3 (Exact tangent certificates). *Over $\mathbb{C}$, $\text{rank } H = 76$ and $\text{rank } J = 545$. Also $JH = 0$ over $\mathbb{Z}$.*

Proof. The checker reconstructs both matrices from the baseline and the monomial exponent matrix. It verifies $JH = 0$ with unbounded integer arithmetic. The file `proofs/laderman_component.json` lists a 76×76 minor of H and a 545×545 minor of J that are nonsingular modulo the prime $p = 1000003$. Primality is checked by trial division, and nonsingularity is checked by exact modular elimination.

An integer determinant nonzero modulo p is a nonzero integer, hence is nonzero over $\mathbb{C}$. Thus $\text{rank } H \geq 76$ and $\text{rank } J \geq 545$. The size of H and the relation $JH = 0$ imply $\text{rank } J \leq 621 - 76 = 545$. All claimed ranks follow. $\square$

Theorem 5.4 (The Laderman component). *Z is the unique irreducible component of X_{23} through L . It has dimension 76, and L is a nonsingular point of X_{23} . In particular, this entire irreducible component has no zero-slot point.*

Proof. The columns of H are derivatives of the action map $G \times S \rightarrow X_{23}$ at (I, L) . Lemma 5.3 gives differential rank 76, so its image closure Z has dimension at least 76. For instance, a nonzero 76-dimensional derivative minor yields local analytic image dimension at least 76 over $\mathbb{C}$.

Every tangent vector to X_{23} at L lies in $\ker J$. Therefore

$$\dim_L X_{23} \leq \dim T_L X_{23} \leq 621 - 545 = 76.$$

Since Z is irreducible and contains L , its dimension gives the reverse lower bound $\dim_L X_{23} \geq \dim Z \geq 76$. Consequently all these dimensions are 76. Equality between local dimension and tangent dimension makes L nonsingular, so exactly one irreducible component of X_{23} passes through L . That component contains Z , and the equal dimensions force it to be Z itself. Theorem 5.2 gives the last assertion. $\square$

Jacobian methods and orbit-dimension bounds for Brent varieties appear in Li–Bao–Zhang [8]. The argument here does not assume that the ideal generated by the Brent polynomials is radical: even if it were not, $\ker J$ would still be an upper bound for the tangent space of its zero set. The 76-dimensional family provides the independent lower bound needed to close the dimension calculation.

5.3 Precisely what the component theorem excludes

It excludes reaching a zero slot by any sequence that stays inside Z , including arbitrary changes of basis and finite limiting operations.

It does not show that $X_{23} = Z$. It does not exclude another irreducible component containing a padded rank-22 algorithm. Nor does it preclude a continuous path that leaves Z through an intersection with another component elsewhere. Finally, all 23 slots being nonzero is not itself proof that the tensor cannot be expressed by a different collection of 22 nonzero slots. These distinctions prevent the local result from being mistaken for $R_{\mathbb{C}}(M_3) = 23$.

6 General reshuffling constraints and the tight case

The next arguments apply to arbitrary exact decompositions, without prescribing support, signs, reality, or a component. Here n may be arbitrary, and all retained slots are assumed nonzero.

Theorem 6.1 (Three matrix-rank-sum inequalities). *For an exact decomposition of $n \times n$ matrix multiplication,*

$$\sum_t \text{rank } U_t \geq n^3, \quad \sum_t \text{rank } V_t \geq n^3, \quad \sum_t \text{rank } W_t \geq n^3. \quad (11)$$

These are ordinary ranks of the coefficient matrices, not tensor ranks.

Proof. Reshape (3) into a matrix whose row is (i, j, k') and column is (i', j', k) . The target becomes I_{n^3} . A summand becomes

$$K_t[(i, j, k'), (i', j', k)] = U_{t,ij} V_{t,j'k} W_{t,i'k'}.$$

Factor $W_t = X_t Y_t^\top$ with $q_t = \text{rank } W_t$ independent columns in each of X_t, Y_t . Then $K_t = L_t R_t$, where

$$L_t[(i, j, k'), a] = U_{t,ij} Y_t[k', a], \quad R_t[a, (i', j', k)] = X_t[i', a] V_{t,j'k}.$$

Thus $\text{rank } K_t \leq q_t$ (indeed equality holds for nonzero U_t, V_t). Subadditivity of matrix rank applied to $I_{n^3} = \sum_t K_t$ gives the W inequality. Cyclically permuting the three factors in the trace form of the multiplication tensor gives the other two inequalities; transposes do not change ordinary matrix ranks. $\square$

For a 22-slot decomposition with $n = 3$, each inequality says $\sum_t (\text{rank } W_t - 1) \geq 5$, and similarly for U and V . This alone is much weaker than a lower bound of 23 products.

Definition 6.2. A decomposition is *output-tight* when $\sum_t q_t = n^3$, where $q_t = \text{rank } W_t$.

Theorem 6.3 (Dual block identities in the tight case). *For an output-tight decomposition, choose any full-column-rank factorizations $W_t = X_t Y_t^\top$. Then for all slots s, t ,*

$$X_s^\top U_t V_s Y_t = \begin{cases} I_{q_t}, & s = t, \\ 0_{q_s \times q_t}, & s \neq t. \end{cases} \quad (12)$$

In particular,

$$q_t \leq \min\{\text{rank } U_t, \text{rank } V_t\}, \quad \sigma_t = q_t. \quad (13)$$

Proof. Concatenate the matrices in the proof of Theorem 6.1: $L = [L_1 \ \cdots \ L_r]$ and $R = [R_1^\top \ \cdots \ R_r^\top]^\top$. The exact identity gives $LR = I_{n^3}$. Tightness makes L, R square, hence $RL = I_{n^3}$ as well. The (s, t) block of RL is

$$\sum_{i,j,k} X_s[i, b] V_s[j, k] U_t[i, j] Y_t[k, a] = (X_s^\top U_t V_s Y_t)_{ba},$$

proving (12). The diagonal blocks imply the rank inequalities in (13). Their traces give $q_t = \text{tr}(X_t^\top U_t V_t Y_t) = \text{tr}(U_t V_t W_t^\top) = \sigma_t$. $\square$

Theorem 6.4 (An invertible output forces 25 slots). *For $n = 3$, an output-tight decomposition with a rank-three output matrix has exactly 25 nonzero slots. Thus an output-tight decomposition with at most 24 nonzero slots has no rank-three output matrix.*

Proof. Let $q_0 = 3$. Then X_0, Y_0 are invertible, and the self-block identity $X_0^\top U_0 V_0 Y_0 = I_3$ makes U_0, V_0 invertible. For any other slot t , the off-diagonal block with $s = 0$ yields

$$U_t V_0 Y_t = 0.$$

The matrix $V_0 Y_t$ has column rank q_t , so $\text{rank } U_t \leq 3 - q_t$. Using the block with second slot index 0 similarly gives

$$X_t^\top U_0 V_t = 0, \quad \text{rank } V_t \leq 3 - q_t.$$

By (13), $q_t \leq \text{rank } U_t, \text{rank } V_t$. Thus $2q_t \leq 3$, and every other $q_t = 1$. Output tightness is now $27 = 3 + (r - 1)$, hence $r = 25$. $\square$

Corollary 6.5 (The output-tight 22-slot profile). *An output-tight 22-slot decomposition of M_3 must have exactly five rank-two outputs and seventeen rank-one outputs. It has no rank-three output. Its contraction signatures have the corresponding multiset $\{2, 2, 2, 2, 2, 1, \dots, 1\}$.*

Proof. By Theorem 6.4, the output ranks lie in $\{1, 2\}$. If a of them equal 2, output tightness gives $27 = 2a + (22 - a)$, so $a = 5$. The signature claim follows from Theorem 6.3. $\square$

No argument here establishes that an arbitrary rank-22 decomposition must be output-tight in any orientation. Therefore Corollary 6.5 does not exclude every possible output profile.

6.1 Application to the Sun baseline

For the supplied Sun data, direct exact matrix-rank computations give

$$\sum_t \text{rank } U_t = 32, \quad \sum_t \text{rank } V_t = 30, \quad \sum_t \text{rank } W_t = 27.$$

The four rank-two outputs are slots 5, 8, 10, 23; all other outputs have rank one. The certificate `proofs/sun_tight.json` gives explicit integer factorizations $W_t = X_t Y_t^\top$. The checker verifies their ranks, forms the 27×27 matrices L, R , checks both $LR = I$ and $RL = I$, and independently verifies all 529 block identities.

Consequently, one cannot delete any Sun slot while requiring the ordinary matrix rank of each remaining output to stay at or below its current value. Such a deletion would lower the output-rank sum below 27, in contradiction to Theorem 6.1, regardless of how the two input factor arrays are changed. A successful reduction would have to escape that output-rank restriction.

7 Verification, reproducibility, and numerical experiments

7.1 Trust boundaries of the certificate checker

The exact checker does not import the certificate generator and does not use a symbolic algebra system or floating-point tolerances. It does share indexing and matrix-building helpers with the generator. Consequently it is a replay checker, not an independent authorship or formal-verification claim. A second, stand-alone checker evaluates arbitrary rational-complex identities directly from six matrix indices and uses only the Python standard library.

The exact operations used are integer arithmetic, integer matrix products, Laurent exponent addition, rational elimination for small coefficient matrices, and finite-field elimination for selected integer minors. Large integer products use unbounded Python integers. Modular operations use 64-bit integer arrays with prime 1000003; every multiplication before reduction is below 10^{12} , within the integer range.

The modular-minor argument is different from transferring a tensor-rank lower bound across fields. A specific nonzero integer determinant remains nonzero over $\mathbb{C}$. A statement that a polynomial system has no solutions with coordinates in $\mathbb{F}_2$ says nothing comparable about its complex solutions.

7.2 Commands

From the root of the extracted archive, with the recorded dependencies installed:

```
python code/verify.py
python code/self_tests.py
python code/verify_candidate.py data/laderman23.json
python code/verify_candidate.py data/sun23.json
```

Regeneration, which additionally uses SymPy, is:

```
python code/build_data.py
python code/generate_certificates.py
python code/verify.py
```

A nontrivial exact rational family member can be created by:

```
python code/instantiate_family.py --parameters 2 3 5 7 11 13 \
  --output results/example.json
python code/verify_candidate.py results/example.json
```

The candidate checker supports $\mathbb{Q}(i)$ coefficients represented by exact rational strings or separate real and imaginary rational parts. It rejects floating-point coefficients. It does not implement arbitrary algebraic number fields, so its accepted coefficient format must not be mistaken for a restriction in the mathematical statement of AC-02.

The seven regression tests include correct real and complex-gauged identities, a nontrivial rational family member, and deliberately corrupted identities, forcing traces, and Laurent certificates. All seven pass. A corruption being rejected is a software regression test, not a replacement for the proof of the verification method.

7.3 Unrestricted numerical exploration

The numerical search frees every coefficient in 22 slots, including entries that were zero in the starting schemes. It is not a support-fixed search. The recorded run uses seed 20260914 and 54 starting points: 46 obtained by deleting each slot of each baseline, plus four random real and four random complex starts. Each receives 500 alternating least-squares sweeps. The four best starts and two complex starts receive up to 30 damped Gauss–Newton steps.

The best recorded point, from deleting Sun slot 21, has

$$\left\| \sum_{t=1}^{22} u_t \otimes v_t \otimes w_t - T \right\|_F = 0.16307567682559754,$$

with relative residual 0.031383928637846385. This is not an exact identity. The file `results/numerical_rank22_search.json` contains all settings and iteration records; the best arrays are retained in `results/numerical_best_rank22.npz`.

No mathematical lower bound follows from these failures. Nor would a very small residual by itself establish an exact rank-22 identity. Coefficients escaping to infinity can represent an approximation phenomenon rather than a finite exact decomposition. The numerical run is reported solely as reproducible exploration and is not used in any proof.

8 The remaining global step

Let

$$\mathcal{N} = \{F \in X_{23} : \text{at least one slot of } F \text{ is zero}\}.$$

By (4), a proof that $\mathcal{N}$ is empty would, together with the verified upper bound, establish $R_{\mathbb{C}}(M_3) = 23$. Theorem 5.4 proves only $Z \cap \mathcal{N} = \emptyset$. No certificate in this archive establishes emptiness on every other component of X_{23} .

There is an equivalent unrestricted polynomial target. Let $I_{22} \subset \mathbb{Q}[U, V, W]$ be the ideal of the 729 Brent polynomials in 594 variables for 22 slots. A global nonexistence proof can be represented by an exact polynomial identity

$$1 = \sum_{a,b,c=0}^8 h_{abc}(U, V, W) \left(\sum_{t=1}^{22} U_{t,a} V_{t,b} W_{t,c} - T_{abc} \right). \quad (14)$$

Such an identity would be a directly checkable certificate of $R_{\mathbb{C}}(M_3) > 22$. Conversely, if the complex zero set is empty, the Nullstellensatz provides an identity over $\mathbb{C}$; since the generators are rational and only finitely many monomial coefficients occur, linear algebra over $\mathbb{Q}$ yields rational h_{abc} as well. No identity of the form (14) is supplied here.

An exact 22-slot construction is the alternative global direction. It would improve the upper bound but would still require a matching lower bound to determine the exact minimum. Restricting such a construction to small integer coefficients, a prescribed support, a symmetric ansatz, or a known component would need an additional completeness theorem before a negative search could resolve AC-02.

8.1 Consequences for further work

The component theorem identifies a concrete unproductive route for a zero-slot reduction: staying within the Laderman component and trying to make a product vanish. Further rank-reduction searches must leave that component or pursue a different decomposition mechanism. The output-tight profile theorem gives a smaller, explicitly conditional branch of a possible classification argument. What is still needed in that branch is a proof that the five rank-two and seventeen rank-one outputs cannot satisfy all the cross-block identities, or a construction showing they can. Neither conclusion is asserted.

The repository's rules require a full resolution of the original field, model, assumptions, and quantifiers for a solved status; a checked special case does not suffice [10]. This archive should therefore be treated as partial research with exact supporting certificates, not as a submission claiming that AC-02 is solved. No repository files were modified.

A The explicit Laderman identity used here

This appendix uses the conventional 1-based matrix indices. Let $A = (a_{ij})$, $B = (b_{ij})$, and define the following products. This is the existing Laderman identity [3], not a new algorithm.

$$\begin{aligned}
p_1 &= (a_{11} + a_{12} + a_{13} - a_{21} - a_{22} - a_{32} - a_{33})(b_{22}) \\
p_2 &= (a_{11} - a_{21})(-b_{12} + b_{22}) \\
p_3 &= (a_{22})(-b_{11} + b_{12} + b_{21} - b_{22} - b_{23} - b_{31} + b_{33}) \\
p_4 &= (-a_{11} + a_{21} + a_{22})(b_{11} - b_{12} + b_{22}) \\
p_5 &= (a_{21} + a_{22})(-b_{11} + b_{12}) \\
p_6 &= (a_{11})(b_{11}) \\
p_7 &= (-a_{11} + a_{31} + a_{32})(b_{11} - b_{13} + b_{23}) \\
p_8 &= (-a_{11} + a_{31})(b_{13} - b_{23}) \\
p_9 &= (a_{31} + a_{32})(-b_{11} + b_{13}) \\
p_{10} &= (a_{11} + a_{12} + a_{13} - a_{22} - a_{23} - a_{31} - a_{32})(b_{23}) \\
p_{11} &= (a_{32})(-b_{11} + b_{13} + b_{21} - b_{22} - b_{23} - b_{31} + b_{32}) \\
p_{12} &= (-a_{13} + a_{32} + a_{33})(b_{22} + b_{31} - b_{32}) \\
p_{13} &= (a_{13} - a_{33})(b_{22} - b_{32}) \\
p_{14} &= (a_{13})(b_{31}) \\
p_{15} &= (a_{32} + a_{33})(-b_{31} + b_{32}) \\
p_{16} &= (-a_{13} + a_{22} + a_{23})(b_{23} + b_{31} - b_{33}) \\
p_{17} &= (a_{13} - a_{23})(b_{23} - b_{33}) \\
p_{18} &= (a_{22} + a_{23})(-b_{31} + b_{33}) \\
p_{19} &= (a_{12})(b_{21}) \\
p_{20} &= (a_{23})(b_{32}) \\
p_{21} &= (a_{21})(b_{13}) \\
p_{22} &= (a_{31})(b_{12}) \\
p_{23} &= (a_{33})(b_{33})
\end{aligned}$$

The output entries are

$$\begin{aligned}
c_{11} &= p_6 + p_{14} + p_{19} \\
c_{12} &= p_1 + p_4 + p_5 + p_6 + p_{12} + p_{14} + p_{15} \\
c_{13} &= p_6 + p_7 + p_9 + p_{10} + p_{14} + p_{16} + p_{18} \\
c_{21} &= p_2 + p_3 + p_4 + p_6 + p_{14} + p_{16} + p_{17} \\
c_{22} &= p_2 + p_4 + p_5 + p_6 + p_{20} \\
c_{23} &= p_{14} + p_{16} + p_{17} + p_{18} + p_{21} \\
c_{31} &= p_6 + p_7 + p_8 + p_{11} + p_{12} + p_{13} + p_{14} \\
c_{32} &= p_{12} + p_{13} + p_{14} + p_{15} + p_{22} \\
c_{33} &= p_6 + p_7 + p_8 + p_9 + p_{23}
\end{aligned}$$

The data ordering, support variety, signature slot labels, and all exact certificates refer to this particular displayed ordering.

B Certificate format and audit map

All JSON label triples (g, t, k) are 0-based: $g = 0, 1, 2$ denotes U, V, W , t is the slot, and k is the row-major coordinate. Restricted variables are enumerated lexicographically over those triples where the baseline entry is nonzero. A Brent polynomial has index $81a + 9b + c$.

Certificate	What must be checked
<code>*_forcing.json</code>	Each referenced equation has two nonzero-coefficient monomials; the designated known monomial is nonzero; the listed new variables are exactly the new factors forced.
<code>*_pairwise.json</code>	In each of three mode pairs, each selected row is supported on a unique remaining column and its two coefficients were forced nonzero.
<code>laderman_torus.json</code>	Initial and final selected minors have integer inverses; pivot formulas and identity free blocks hold; integer row combinations prove four cancellations; all 729 Laurent residuals vanish.
<code>laderman_gauge_slice.json</code>	The gauge exponent matrix is reconstructed; $ED = B$; the selected gauge minor is unimodular; the six remaining parameter indices and exponent columns agree.
<code>laderman_component.json</code>	$JH = 0$ over the integers; the specified 545- and 76-dimensional minors are nonsingular modulo the verified prime.
<code>sun_tight.json</code>	Every output factorization and its rank is exact; $LR = RL = I_{27}$; all 529 cross-block identities hold.

No hidden oracle. The generator may use SymPy to find inverse matrices and decompositions, but the checker verifies the resulting finite identities directly. The correctness argument for the component theorem additionally uses the explicitly written algebraic-geometric reasoning in Section 5; a successful Python run alone is not a formal verification of that reasoning.

Integrity and reproducibility. MANIFEST.sha256 records the files in the distributed archive. The environment versions and exact run commands are in `environment.json` and `README.md`. Historical source papers are cited rather than bundled. Numeric residuals can vary slightly with numerical linear-algebra libraries, whereas the exact verification assertions must have the same truth value.

References

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